

B5. Bioenergetics

Multiple choice questions.

- | | | Obj/
Act. Verb/
Diff. Level |
|----|---|--|
| 1. | <p>Read and analyse the following statements:</p> <p>Statement 1 : Water is the electron donor for non-cyclic photo phosphorylation.</p> <p>Statement 2 : Both ATP and NADPH₂ are synthesized during cyclic photophosphorylation.</p> <p>a) Both 1 and 2 are correct.
 b) Both 1 and 2 are wrong.
 c) 1 is correct and 2 is wrong.
 d) 1 is wrong and 2 is correct.</p> | A
Analyses
Difficult |
| 2. | <p>Which is the ultimate source of energy for all living organisms?</p> <p>a) Food
 b) Water
 c) Solar energy
 d) Chemical energy</p> | K
Recognizes
Average |
| 3. | <p>Read and analyse the following statements.</p> <p>Statement 1 : One of the end products of photosynthesis is the raw material for respiration.</p> <p>Statement 2 : One of the end products of respiration is the raw material for photosynthesis.</p> <p>a) Both 1 and 2 are correct.
 b) Both 1 and 2 are wrong.
 c) 1 is correct and 2 is wrong.
 d) 1 is wrong and 2 is correct.</p> | A
Analyses
Difficult |
| 4. | <p>Identify the correct sequence of electron carriers in cyclic photophosphorylation.</p> <p>a) P₇₀₀ - Fe-S → PC → Cyt b → Cyt f
 b) P₇₀₀ - Fe-S → cyt b → cyt f → PC
 c) P₇₀₀ - Fe-S → cyt b → cyt f → PQ
 d) P₇₀₀ - Fe-S → Fe-S → cyt b → PC</p> | U
Sequences
Difficult |
| 5. | <p>Identify the correct combination of colours of light that are responsible for Emerson's enhancement effect.</p> <p>a) Red and blue
 b) Red and green
 c) Blue and green
 d) Blue and orange</p> | U
Identifies
Difficult |

6. Which is the acceptor of CO_2 in Calvin cycle?
 a) PEP
 b) RUMP
 c) RUBP
 d) PGA
 U
 Identifies
 Average
7. Identify the five carbon compound that absorbs CO_2 in Calvin cycle.
 a) Oxaloacetic acid
 b) Malic acid
 c) Aspartic acid
 d) Ribulosebiphosphate
 U
 Identifies
 Average
8. In Calvin cycle, theoretically the number of molecules of CO_2 required for the synthesis of one molecule of glucose is
 a) 2
 b) 4
 c) 6
 d) 8
 U
 Correlates
 Average
9. Identify the incorrect statement.
 a) Xanthophylls are accessory pigments
 b) Light reaction takes place in the stroma of chloroplasts,
 c) Only ATP molecules are produced during cyclic photophosphorylation
 d) Six molecules of CO_2 are necessary to produce one molecule of glucose.
 U
 Identifies
 Difficult
10. Match the phenomena in column I with their sites of occurrence in column II.
- | Column I | Column II |
|------------------------------------|-------------------------|
| i) Krebs cycle | p) cytoplasm |
| ii) Synthesis of ATP | q) grana |
| iii) Glycolysis | r) $F_0 - F_1$ particle |
| iv) Light reaction | s) mitochondrial matrix |
| a) i - s; ii - r; iii - p; iv - q | |
| b) i - r; ii - p; iii - q; iv - s | |
| c) i - p; iii - q; iii - s; iv - r | |
| d) i - s; ii - p; iii - r; iv - q | |
- U
 Identifies
 Average
11. Choose the correct statement from the following :
 a) In CAM plants stomata open during day time.
 b) Sugarcane plant exhibits C_4 pathway.
 c) Chlorophyll pigments absorb maximum light in the green wavelength.
 d) Reaction centre of PS - II is P_{700} .
 U
 Identifies
 Difficult

- | | |
|---|---|
| <p>12. Identify the mismatch.</p> <p>a) PSI – P₇₀₀</p> <p>b) Glycolysis – Pyruvic acid</p> <p>c) C₄ plants – Krantz anatomy</p> <p>d) Emerson – Glycolysis</p> | <p>U</p> <p>Identifies</p> <p>Difficult</p> |
| <p>13. Identify the type of reaction that occurs when Oxalosuccinic acid is converted to α-Ketoglutaric acid in the Krebs cycle.</p> <p>a) Oxidation</p> <p>b) Reduction</p> <p>c) Decarboxylation</p> <p>d) Phosphorylation</p> | <p>U</p> <p>Identifies</p> <p>Difficult</p> |
| <p>14. Identify the types of reaction that occurs during the conversion of pyruvic acid to acetyl CoA.</p> <p>a) Oxidation and reduction</p> <p>b) Hydration and carboxylation</p> <p>c) Dehydrogenation and decarboxylation</p> <p>d) Decarboxylation and hydrogenation</p> | <p>U</p> <p>Explains</p> <p>Difficult</p> |
| <p>15. In photosynthesis Warburg's effect explains that</p> <p>a) increased conc. of CO₂ reduces photosynthesis</p> <p>b) increased conc. of O₂ reduces photosynthesis</p> <p>c) increased conc. of CO₂ increases photosynthesis</p> <p>d) increased conc. of O₂ increases photosynthesis</p> | <p>U</p> <p>Interprets</p> <p>Difficult</p> |
| <p>16. How many molecules of glucose are required to produce 36 ATPs in anaerobic respiration by yeast cell?</p> <p>a) 36</p> <p>b) 18</p> <p>c) 9</p> <p>d) 1</p> | <p>A</p> <p>Analyses</p> <p>Difficult</p> |
| <p>17. In which stage of Krebs cycle FAD is reduced to FADH₂?</p> <p>a) α - Ketoglutaric acid to succinyl CoA.</p> <p>b) Malic acid to oxaloacetic acid</p> <p>c) Succinic acid to fumaric acid</p> <p>d) Iso-citric acid to oxalo succinic acid</p> | <p>U</p> <p>Identifies</p> <p>Difficult</p> |
| <p>18. If the RQ value is more than unity (one), the respiratory substrate is</p> <p>a) carbohydrate</p> <p>b) protein</p> <p>c) fat</p> <p>d) organic acid</p> | <p>A</p> <p>Analyses</p> <p>Average</p> |

19. How many molecules of ATP are used up during glycolysis?
 a) 2
 b) 6
 c) 8
 d) 10
 K
 Recalls
 Easy
20. The $F_0 - F_1$ particles are found on
 a) matrix side of inner mitochondrial membrane
 b) inter membrane of inner mitochondrial membrane
 c) inter membrane space side of outer mitochondrial membrane
 d) the outer surface of outer mitochondrial membrane
 U
 Identifies
 Average
21. Total number of ATP produced by the oxidation of one molecule of Maltose is
 a) 36
 b) 72
 c) 38
 d) 76
 S
 Solves
 Difficult
22. The RQ value of which one of the following is unity (one)?
 a) Casein
 b) Fructose
 c) Albumin
 d) Wax
 A
 Analyses
 Difficult
23. Number of O_2 molecules required for the oxidation of one molecule of glucose during aerobic respiration is
 a) 2
 b) 4
 c) 6
 d) 8
 A
 Analyses
 Difficult

Very short answer questions

24. Name the enzyme that catalyzes the carboxylation step in Calvin cycle.
 U
 Identifies
 Average
25. Which is the primary source of energy for all living organism?
 K
 Recalls
 Easy
26. What are quantasomes?
 K
 Defines
 Average

27.	What are photosystems / pigment systems?	K Defines Average
28.	What is a photon?	K Defines Average
29.	What is a quantum?	K Defines Average
30.	Define photoionization of water.	K Defines Average
31.	Name the gas that is liberated as a byproduct during photosynthesis.	K Recalls Easy
32.	Name the molecule that absorbs CO ₂ during C ₃ pathway.	U Identifies Average
33.	Name the enzyme that catalyses carboxylation process during C ₄ pathway.	K Recalls Average
34.	Name the molecule that absorbs CO ₂ during C ₄ pathway.	K Recalls Average
35.	Name the electron donor for non-cyclic photophosphorylation.	K Recalls Average
36.	Name the first stable intermediate compound formed during C ₄ pathway.	U Identifies Average
37.	Name the first stable intermediate compound formed during C ₃ pathway.	U Identifies Average
38.	Why Calvin cycle is also known as C ₃ pathway?	U Explains Average
39.	Why Hatch and Slack cycle is also known as C ₄ pathway?	U Identifies Average
40.	What is solarization?	K Defines Difficult

41. What is Emerson's effect?	K Defines Difficult
42. Define Blackman's law of limiting factor.	K Defines Average
43. What is Warburg's effect?	K Defines Difficult
44. In the chloroplast, name the site of light reaction of photosynthesis in chloroplast.	K Recalls Easy
45. Name the plant group adapted for the CAM pathway of carbon fixation.	K Recalls Easy
46. Write the balanced equation of photosynthesis.	U Formulates Average
47. Expand the abbreviation CAM.	K Recalls Easy
48. Expand the term RUBISCO.	K Recalls Easy
49. Expand the abbreviation NAD.	K Recalls Average
50. Define photosynthesis.	K Defines Easy
51. Define respiration.	K Defines Easy
52. Which cell organelle is associated with aerobic respiration?	K Recalls Easy
53. What is aerobic respiration?	K Defines Easy
54. Where does glycolysis occur in cells?	K Recalls Easy

55.	What is anaerobic respiration?	K Defines Easy
56.	Write the chemical equation of aerobic respiration with glucose as the substrate.	K Recalls Average
57.	What is fermentation?	K Defines Easy
58.	Within the mitochondrion, where exactly does Krebs cycle occur?	K Recalls Easy
59.	What are anaerobes?	K Defines Easy
60.	How many glucose molecules are required to produce 36 ATP molecules by an yeast cell through anaerobic respiration?	A Analyses Difficult
61.	Define Pasteur effect.	K Defines Difficult
62.	Name the site of electron transport system in mitochondria.	K Recalls Average
63.	What are facultative anaerobes?	K Defines Easy
64.	Expand the term FAD.	K Recalls Easy
65.	What are obligatory anaerobes?	K Defines Easy
66.	Which gas is released during respiration in plants?	K Recalls Easy
67.	Define respiratory quotient.	K Defines Average
68.	Write the equation of lactate fermentation in lactic acid bacteria.	K Recalls Average

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|--|-----------------------------|
| 69. What is oxidative phosphorylation? | K
Defines
Average |
| 70. Why is the RQ of proteins always less than one? | A
Analyse
Average |
| 71. What is terminal oxidation? | K
Defines
Average |
| 72. How many molecules of ATP are produced by the oxidation of one FADH ₂ molecule? | K
Recalls
Average |
| 73. How many molecules of ATP are produced by the oxidation of one NADH ₂ molecule? | K
Recalls
Easy |
| 74. Expand the term ATP. | K
Recalls
Easy |
| 75. Glycolysis does not require mitochondria. Give reason. | A
Reasons out
Average |
| 76. Name the compound which links glycolysis and Krebs cycle. | U
Identifies
Average |
| 77. Why does a glucose molecule generate only two ATP molecules in anaerobic respiration. | A
Reasons out
Average |

Short answer questions

- | | |
|---|--------------------------------------|
| 78. Write the schematic representation of cyclic photophosphorylation. | S
Sequences
Average |
| 80. Sketch and label the ultrastructure of a stoma. | S, U
Draws,
Identifies
Easy |
| 81. Mention any four environmental factors which affect the rate of photosynthesis. | K
Recalls
Easy |
| 82. What are accessory photosynthetic pigments? Name two accessory pigments. | U
Explains
Difficult |

- | | |
|--|----------------------------|
| 84. Name the end products of non-cyclic photophosphorylation. | U
Identifies
Average |
| 86. What is meant by Kranz anatomy? Mention one example. | U
Explains
Difficult |
| 87. Write the substance that absorbs CO ₂ in C ₄ pathway and the enzyme that catalyses its absorption. | U
Explains
Average |
| 90. Draw a labeled diagram of a mitochondrion. | S
Draws
Easy |
| 94. Mention the external factors affecting respiration in plants. | K
Mentions
Easy |

Long answer questions

- | | |
|---|--|
| 96. Write the schematic representation of cyclic photophosphorylation. Show the path of electrons and the site of ATP generation. | U
Sequences
Average |
| 97. Define and write the schematic representation of non-cyclic photophosphorylation. Indicate the site of photolysis, photosystems, ATP and NADPH ₂ generation. | S
Sequences,
explains
Average |
| 98. Schematically represent Calvin cycle. Explain the three important phases (stages). | U
Sequences,
explains
Difficult |
| 99. Name the factors that affect the rate of photosynthesis. Explain the role of any three factors. | U
Explains
Easy |
| 100. With the help of a neat labeled diagram, explain the ultrastructure of chloroplast. | S
Draws
Easy |
| 101. List five differences between aerobic and anaerobic respiration. | U
Differentiates
Average |
| 102. List five differences between respiration and photosynthesis. | U
Differentiates
Average |
| 103. What is fermentation? Explain the steps during alcoholic fermentation. Write the equation of fermentation. | U
Explains
Average |

- | | | |
|------|--|-----------------------------|
| 104. | List four external factors which affect the rate of respiration.
Explain the role of each factor. | U
Explains
Easy |
| 105. | Write the schematic representation of Krebs cycle. Mention the names of intermediates, enzymes and co-enzymes involved in the process. | S
Sequences
Difficult |
| 106. | Give a schematic representation of electron transport system. | S
Sequences
Difficult |
| 107. | Write a flow chart of glycolysis indicating all the intermediates, enzymes and co-enzymes. | S
Sequences
Difficult |
| 108. | Describe the ultrastructure of mitochondrion with a neat labeled diagram. | S
Draws
Easy |

ANSWERS

Multiple choice - answers (one mark each)

1. c
2. c
3. a
4. b
5. a
6. c
7. d
8. c
9. b
10. a
11. b
12. d
13. c
14. c
15. b
16. b
17. c
18. d
19. a
20. a
21. b
22. b
23. c

Very short answers (one mark each)

24. RUBP carboxylase / oxygenase or RUBISCO.
25. Sunlight
26. These are the structural and functional units present on the membrane of thylakoids which harvest light energy during photosynthesis.
27. These are the photosynthetic pigment assemblies present in the thylakoid membrane of chloroplasts.
28. Photon is the particle of light containing energy.
29. The amount of energy present in a photon is called quantum.

30. It is the process of splitting of water into H^+ and OH^- and electrons in the presence of light energy.
31. Oxygen (O_2)
32. RUBP (Ribulose biphosphate)
33. Phosphoenol pyruvic acid carboxylase. (PEP carboxylase)
34. Phosphoenol pyruvic acid (PEP)
35. Water (H_2O)
36. Oxalolacetic acid (OAA) / Malic acid / aspartic acid.
37. Phosphoglyceric acid or PGA.
38. The first stable compound produced during Calvin cycle is the PGA which is a 3C compound.
39. It is called C_4 pathway because the first stable compound produced is OAA which is a 4C compound.
40. It is the photodestruction (bleaching) of chlorophyll pigments under the influence of high intensity sunlight.
41. The simultaneous supply of red and blue colour of light will enhance the rate of photosynthesis as compared to the sum of the rates of photosynthesis when the two wavelengths of light are given separately.
42. When a process is conditioned as to its rapidity by several factors, the rate of the process is limited by the pace of the slowest factor.
43. Increased concentration of O_2 in the mesophyll tissue reduces the rate of photosynthesis.
44. Grana of the chloroplast / photosystems / thylakoid.
45. Succulents / Crassulaceae members.
46.
$$6CO_2 + 12 H_2O \xrightarrow[\text{Chlorophyll}]{\text{Light}} C_6H_{12}O_6 + 6H_2O + 6O_2$$
47. Crassulacean acid metabolism.
48. Ribulose biphosphate carboxylase/ oxygenase.

49. Nicotinamide adenine dinucleotide.
50. Synthesis of organic molecules like glucose by utilizing inorganic substances like CO₂ and H₂O in the presence of sunlight and chlorophyll pigments.
51. Respiration is the process of oxidation of food substances (with or without the need of oxygen) for liberation of energy.
52. Mitochondria
53. The breakdown (oxidation) of food substances by organisms in the presence of oxygen is called aerobic respiration.
54. Cytoplasm of cell.
55. The breakdown (oxidation) of food substances in organisms in the absence of oxygen is called anaerobic respiration.
56.
$$\text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2 \longrightarrow 6\text{CO}_2 + 6\text{H}_2\text{O} + 36 \text{ ATP (energy)}$$
57. Fermentation is a type of anaerobic respiration resulting in the formation of alcohol, CO₂ and energy.
58. Mitochondrial matrix
59. The organisms which do not require oxygen for respiration are called anaerobes.
60. 18 molecules of glucose.
61. The shift from anaerobic respiration to aerobic respiration in the presence of oxygen by certain facultative anaerobes like yeast is called Pasteur effect.
62. Inner mitochondrial membrane
63. Anaerobes which live, preferentially, in the presence of oxygen, but which can also live in the absence of oxygen are facultative anaerobes.
64. Flavin Adenine Dinucleotide.
65. The anaerobes which survive only in the absence of oxygen are called obligatory anaerobes.
66. Carbon dioxide

67. The ratio of amount of carbon dioxide liberated to the amount of oxygen consumed during respiration is called respiratory quotient.
68.
$$\text{C}_6\text{H}_{12}\text{O}_6 \longrightarrow 2\text{C}_3\text{H}_6\text{O}_3 + 2\text{ATP}$$

GlucoseLactate
69. Synthesis of ATP from ADP using the energy produced during oxidation of NADH_2 and FADH_2 .
70. Since the C to O ratio in proteins is always higher for C, it requires more O_2 for breakdown. Hence the RQ is below one.
71. The oxidation of reduced coenzymes like NADH_2 and FADH_2 through the ETS of mitochondria is called terminal oxidation.
72. Two
73. Three
74. Adenosine triphosphate
75. All the enzymes of glycolysis are situated in the cytoplasm.
76. Acetyl CoA
77. Glycolysis yields only 2 ATP molecules and the products of glycolysis do not undergo any further oxidation during anaerobic respiration.

Short answers

78. Schematic representation mentioning PS – I, FRS, FD, cyt b6, cyt f, plasto cyanin and formation of ATP.
(Schematic diagram – 1; Intermediates (at least 4) – 1)
80. Diagram showing parts like guard cells, subsidiary cells, stoma, nucleus, cytoplasm
($\frac{1}{2} \times 4 = 2$ marks)
81. Any four of the following factors: light, temperature, conc. of CO_2 , conc. of O_2 , water ($\frac{1}{2} \times 4$ marks)
82. Accessory pigments can only harvest light energy and transfer to primary pigments. Carotenoids, Xanthophylls are examples of accessory pigments.
(Definition –1 mark, examples : $\frac{1}{2} + \frac{1}{2}$)

84. ATP and NADPH₂
(1 + 1 marks)
86.
 - Presence of a ring of bundle-sheath cells around the vascular bundle.
 - Grass/ sugarcane/ Monochlamydeae
(1 + 1 = 2 marks)
87. Phospho-enolpyruvate; PEP carboxylase
(1 + 1 = 2 marks)
90. Neat diagram with any four labels : cristae, matrix, inner membrane, outer membrane, F₀ – F₁ particles, DNA, ribosomes
(diagram –1, labels $\frac{1}{4} \times 4 - 1 = 2$ marks)
94. Any four of the following factors : Light, temperature, wind, humidity, O₂ concentration.

Long answers

96. Complete schematic representation showing flow chart of electrons (3 marks).
Explanation : Path of electrons (1 mark), site of ATP synthesis (1 mark)
97. Schematic representation - 3 marks
Correct representation of electron carriers – 1
Definition – 1
98. Schematic representation with all intermediates and enzymes.
Explanation of
 - Carboxylation
 - Reduction and glucose synthesis
 - Regeneration
(Schematic representation –3, explanation of 3 phases – 2 = 5 marks)
99. Any four of the following factors.
 - Light
 - Temperature
 - Conc. Of CO₂
 - Conc. Of O₂
 - Availability of H₂O
Explaining any three factors
(Naming – 2, explaining $1 \times 3 - 3 = 5$ marks)
100. Labeled diagram showing thylakoid, grana stroma, fret channel, outer and inner membrane, inter membrane space.
(Diagram –2, labels (any four) – 1, explanation – 2 = 5 marks)

101. **Aerobic respiration**
- O₂ is required.
 - Glucose undergoes complete oxidation.
 - CO₂ and H₂O are the byproducts.
 - Six molecules of CO₂ are liberated.
 - Krebs cycle and electron transport system takes place.
 - 36 ATPs are produced.
 - Occurs in cytoplasm and mitochondria.
- Anaerobic respiration**
- O₂ is not required.
 - Incomplete oxidation.
 - Ethyl alcohol, O₂, lactic acid and CO₂
 - Less than six molecules of CO₂ liberated.
 - No Krebs cycle and electron transport system.
 - 2 ATPs are produced.
 - Occurs in cytoplasm.

(Each difference – 1 mark. Total = 5 marks)

102. Any five of the following differences.

- | Respiration | Photosynthesis |
|---|---|
| • Occurs in all living organisms. | • Occurs in only green plants. |
| • Organelle involved is mitochondria. | • Organelle involved is chloroplast. |
| • Glucose is oxidized to yield H ₂ O and CO ₂ and energy. | • H ₂ O and CO ₂ are used to synthesis glucose. |
| • Catabolic process. | • Anabolic process. |
| • Light is not a requirement. | • Light is a requirement |
| • CO ₂ is released. | • O ₂ is released. |

(Each difference × 1 = 5 marks)

103. Fermentation is a type of anaerobic respiration resulting in the formation of alcohol, CO₂ and energy.

(Definition – 1, equation – 2, explanation -2 = 5 marks)

104. Any four of the following factors :

- Light
- Temperature
- Carbon dioxide
- Oxygen
- Humidity
- Wind

(Mentioning 4 factors – 2 marks. Explaining (1 + 1 + ½ + ½) – 3 marks)

105. Correct sequential schematic representation with all intermediate compounds, enzymes and co-enzymes. (5 marks)

106. Correct sequential schematic representation with all intermediate compounds and co-enzymes. (5 marks)
107. Correct sequential schematic representation with all intermediate compounds and co-enzymes. (5 marks)
108. Neat diagram with any four labels.
- outer membrane
 - inner membrane
 - cristae
 - matrix
 - $F_0 - F_1$ particles
- (Diagram -2, labels (any four) -1, explanation -2 = 5 marks)

